

LINEAR SEARCH:

```
import java.util.*;
public class Main
{
    public static void main(String[] args)
    {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter the size of the array: ");
        int n = sc.nextInt();
        int[] arr = new int[n];
        System.out.println("Enter the array elements:");
        for (int i = 0; i < n; i++)
        {
            arr[i] = sc.nextInt();
        }
        System.out.print("Enter the element to search: ");
        int key = sc.nextInt();
        for (int i = 0; i < n; i++)
        {
            if (arr[i] == key)
            {
                System.out.println("Element found at index: " + i);
                return;
            }
        }
        System.out.println("Element not found");
    }
}
```

OUTPUT:

```
Enter the size of the array: 6
Enter the array elements:
9 2 4 7 8 1
Enter the element to search: 8
Element found at index: 4
```

BINARY SEARCH:

```
import java.util.*;
public class Main
{
    public static void main(String[] args)
    {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter the size of the array: ");
```

```

int n = sc.nextInt();
int[] arr = new int[n];
System.out.println("Enter the array elements (Sorted):");
for (int i = 0; i < n; i++)
{
    arr[i] = sc.nextInt();
}
System.out.print("Enter the element to search: ");
int key = sc.nextInt();
int low = 0;
int high = n - 1;
while (low <= high)
{
    int mid = (low + high) / 2;
    if (arr[mid] == key)
    {
        System.out.println("Element found at index: " + mid);
        return;
    }
    else if (key < arr[mid])
    {
        high = mid - 1;
    }
    else
    {
        low = mid + 1;
    }
}
System.out.println("Element not found");
}
}

```

OUTPUT:

```

Enter the size of the array: 6
Enter the array elements (Sorted):
1 2 3 4 5 6
Enter the element to search: 6
Element found at index: 5

```

INSERTION IN A SORTED ARRAY

```

import java.util.*;
public class Main
{
    public static void main(String[] args)

```

```

{
    Scanner sc = new Scanner(System.in);
    System.out.print("Enter the size of the array: ");
    int n = sc.nextInt();
    int[] arr = new int[n];
    System.out.println("Enter the array elements (Sorted):");
    for (int i = 0; i < n; i++)
    {
        arr[i] = sc.nextInt();
    }
    System.out.print("Enter the element to search: ");
    int key = sc.nextInt();
    int low = 0;
    int high = n - 1;
    while (low <= high)
    {
        int mid = (low + high) / 2;
        if (arr[mid] == key)
        {
            System.out.println("Element found at index: " + mid);
            return;
        }
        else if (key < arr[mid])
        {
            high = mid - 1;
        }
        else
        {
            low = mid + 1;
        }
    }
    System.out.println("The Element Should be inserted at index: " + low);
}
}

```

OUTPUT:

```

Enter the size of the array: 6
Enter the array elements (Sorted):
1 2 3 4 5 7
Enter the element to search: 6
The Element Should be inserted at index: 5

```

FIND THE PEAK ELEMENT:

```
import java.util.*;
```

```

public class Main
{
    public static void main(String[] args)
    {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter the size of the array: ");
        int n = sc.nextInt();
        int[] arr = new int[n];
        System.out.println("Enter the rotated sorted array:");
        for (int i = 0; i < n; i++)
        {
            arr[i] = sc.nextInt();
        }
        int low = 0;
        int high = n - 1;
        if (arr[low] <= arr[high])
        {
            System.out.println("Peak Element: " + arr[high]);
            System.out.println("Index: " + high);
            return;
        }
        while (low <= high)
        {
            int mid = low + (high - low) / 2;
            if (mid < n - 1 && arr[mid] > arr[mid + 1])
            {
                System.out.println("Peak Element: " + arr[mid]);
                System.out.println("Index: " + mid);
                return;
            }
            if (arr[mid] >= arr[low])
            {
                low = mid + 1;
            }
            else
            {
                high = mid - 1;
            }
        }
    }
}

```

OUTPUT:

Enter the size of the array: 7

Enter the rotated sorted array:

4 5 6 7 0 1 2

Peak Element: 7

Index: 3

BUBBLE SORT:

```
import java.util.*;
public class Main
{
    public static void main(String[] args)
    {
        int[] arr = {5, 1, 4, 2, 8};
        int n = arr.length;
        for (int i = 0; i < n - 1; i++)
        {
            for (int j = 0; j < n - i - 1; j++)
            {
                if (arr[j] > arr[j + 1])
                {
                    int temp = arr[j];
                    arr[j] = arr[j + 1];
                    arr[j + 1] = temp;
                }
            }
        }
        System.out.println("Sorted Array:");
        for (int x : arr)
        {
            System.out.print(x + " ");
        }
    }
}
```

OUTPUT:

Sorted Array:

1 2 4 5 8

SELECTION SORT:

```
import java.util.*;
public class Main
{
    public static void main(String[] args)
    {
        int[] arr = {64, 25, 12, 22, 11};
```

```

int n = arr.length;
for (int i = 0; i < n - 1; i++)
{
    int minIndex = i;
    for (int j = i + 1; j < n; j++)
    {
        if (arr[j] < arr[minIndex])
        {
            minIndex = j;
        }
    }
    int temp = arr[i];
    arr[i] = arr[minIndex];
    arr[minIndex] = temp;
}
System.out.println("Sorted Array:");
for (int x : arr)
{
    System.out.print(x + " ");
}
}
}

```

OUTPUT:

Sorted Array:
11 12 22 25 64

INSERTION SORT:

```

import java.util.*;
public class Main
{
    public static void main(String[] args)
    {
        int[] arr = {9, 5, 1, 4, 3};
        int n = arr.length;
        for (int i = 1; i < n; i++)
        {
            int key = arr[i];
            int j = i - 1;
            while (j >= 0 && arr[j] > key)
            {

                arr[j + 1] = arr[j];
                j--;
            }
        }
    }
}

```

```
    }  
    arr[j + 1] = key;  
  }  
  System.out.println("Sorted Array:");  
  for (int x : arr)  
  {  
    System.out.print(x + " ");  
  }  
}  
}
```

OUTPUT:

Sorted Array:

1 3 4 5 9